

Hardware Integrity Expert System

System Execution & Database Verification Report

TARGET SYSTEM hardware_integrity_es.py	WORKSPACE PATH Expert Systems / AI-Training	EVALUATION DATE 23 September 2026
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1. Executive Summary

This report documents the verification process and execution results for the **Hardware Integrity Expert System (ES)**. The expert system is designed to validate hardware component records against system configuration state stored in the database. This evaluation compares two sequential operational turns: the initial diagnostic run identifying database inconsistencies, and the subsequent verification turn following database remediation.

2. Turn 1: Initial Database Inconsistencies

During the initial execution of hardware_integrity_es.py, the expert system evaluated the active database records and reported state non-conformance. The rule engine flagged two distinct hardware discrepancies related to memory configuration and disk identification parameters.

Diagnostic Result: System Not Consistent

The expert system successfully intercepted configuration drift and isolated non-matching parameters between hardware inventory logs and baseline database entries.

```
C:\Users\pc\PycharmProjects\AI-Training\.venv\Scripts\python.exe
"C:\Users\pc\PycharmProjects\AI-Training\Expert Systems\hardware_integrity_es.py" The
computer is not consistent Reasons below Memory partNumber does not match. Disk
serialNumber does not match. Process finished with exit code 0
```

Identified Rule Violations:

Component	Parameter Checked	Status / Finding
Memory	partNumber	Mismatch Detected
Disk	serialNumber	Mismatch Detected

3. Database Remediation Actions

To resolve the integrity failures identified in Turn 1, corrective updates were applied to the database records (leveraging localized scripts such as `changes.sql`). These updates synchronized the stored hardware baseline values with the physical system components.

Specifically, the following database corrections were implemented:

- **Memory Module:** Updated the recorded `partNumber` attribute to accurately reflect installed RAM specifications.
- **Storage Disk:** Corrected the `serialNumber` string field in the storage inventory table to match the active disk hardware identifier.

4. Turn 2: Post-Fix Verification Run

Following database adjustments, the expert system script was re-executed to validate compliance. The inference engine successfully re-evaluated all rule sets against the updated database state.

Diagnostic Result: System Consistent

All hardware integrity checks passed successfully with zero rule violations. The database records are fully synchronized with expected hardware parameters.

```
C:\Users\pc\PycharmProjects\AI-Training\.venv\Scripts\python.exe
"C:\Users\pc\PycharmProjects\AI-Training\Expert Systems\hardware_integrity_es.py" The
computer is consistent Process finished with exit code 0
```

5. Anti-Counterfeit & Supply Chain Applications

Beyond internal auditing, this expert system architecture provides critical utility for **Original Equipment Manufacturers (OEMs) and constructors to detect counterfeit hardware and component substitution:**

- **Component Spoofing Detection:** Automatically flags remarketed parts, fake RAM part numbers, or altered disk serial numbers introduced during supply chain transit.
- **Warranty & RMA Protection:** Prevents component-swapping fraud during returns by verifying that returned devices maintain their exact original bill of materials.